Executive Summary

A/B Test Analysis for Online Shooter Game:

1. Project Objective

The goal of this study was to analyze an A/B test for a shooter game. We offered a **discount on premium armor** to a "Test" group of players. I had to determine if this promotion actually increased revenue and if the results were statistically significant.

2. Data Preparation and Cleaning

The raw data was large and complex (over 8.6 million rows). My first task was to build a clean foundation for analysis: **Deduplication**, **Handling Outliers, Data Merging.**

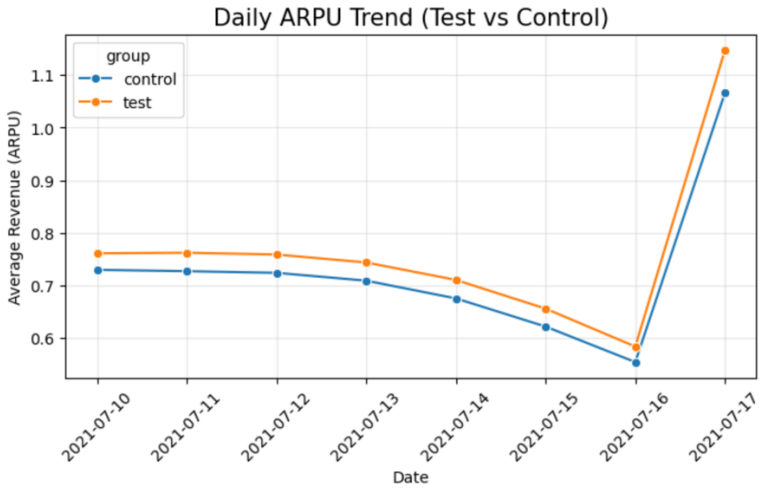
3. Methodology

I used a structural approach (PACE framework) and the following methods:

- **EDA (Exploratory Data Analysis):** Investigating distributions and platform-specific behavior.
- **Statistical Testing:** I used **Welch's T-test** and **95% Confidence Intervals** to validate the difference between groups.
- **Machine Learning (Clustering):** I implemented **K-Means clustering** to segment players into "Casual Spenders" and "High-Value Players" based on their behavior.

4. Key Findings

- **Revenue Growth:** The ARPU (Average Revenue Per User) increased from **$0.73 to $0.76**. This is a **4.1% uplift**.
- **Statistical Significance:** The p-value was extremely low (close to zero), and the Confidence Intervals for the two groups did not overlap. This proves the result was **not due to luck**.
- **Consistency:** Daily charts showed that the Test group performed better than the Control group every single day of the test.
- **Segmentation Insight:** Clustering revealed that the discount worked for all segments, but "Casual" players were slightly more sensitive to the price change.



5. Final Recommendations

Launch Globally: The armor discount is successful and safe. I recommend rolling it out to all players.

Targeting: Future discounts should focus on "Casual" players to encourage them to become active spenders.

Security Update: I suggest using the $5,100 spending threshold to automatically flag potential new cheaters in the future.